

Statistical Analysis of Sepsis-3 in Critically Ill Patients (MIMIC-IV)

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Table of Contents

1	Introduction	2
2	Exploratory Data Analysis	3
3	Hypothesis Testing	4
3.1	Test 1: Maximum lactate (Welch t-test)	4
3.2	Test 2: Maximum white-cell count (Wilcoxon rank-sum)	5
3.3	Test 3: Tachypnoea and sepsis (Chi-squared)	5
4	Maximum Likelihood Estimation	6
5	Generalised Linear Models	6
5.1	Logistic Regression Coefficients	7
5.2	Model Comparison	7
6	Density Estimation and Mixture Models	8
6.1	Kernel Density Estimate	8
6.2	Two-component Gaussian mixture	9
7	Bayesian Inference and MCMC	9
7.1	Part A: Bayesian inference via Metropolis-Hastings	9
7.1.1	Model setup	9
7.1.2	Likelihood and posterior	10
7.1.3	Convergence diagnostics	10
7.1.4	Posterior summary and comparison with MLE	10
7.2	Part B: Hidden Markov Model	11
7.2.1	Choosing the number of states	11
7.2.2	Emissions and transition matrix	11
7.2.3	Unsupervised classification accuracy	12
8	Predictive Validation, Calibration and Missingness	12
8.1	Out-of-sample discrimination	12
8.2	Calibration and clinical usefulness	12
8.3	Non-linearity of the lactate effect	14
8.4	Multicollinearity	15
8.5	The missing-data mechanism	15

9 Conclusion	16
Acknowledgment	17
References	17

1 Introduction

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection, and it remains one of the leading causes of death in intensive care units worldwide. The **Sepsis-3** consensus (Singer et al., 2016) defines sepsis operationally as a **suspected infection** accompanied by an acute rise of **two or more points** in the Sequential Organ Failure Assessment (**SOFA**) score.

This report analyses **MIMIC-IV v3.1**, a large, de-identified database of patients admitted to the critical-care units of the Beth Israel Deaconess Medical Center. From the raw tables a reproducible pipeline (scripts 01 to 05) builds an adult, first-ICU-stay cohort, detects suspected infection from antibiotic orders paired with cultures, computes the six SOFA organ scores from the first 24 hours of each stay, and applies the Sepsis-3 rule. The result is an analysis matrix of **65,241** ICU stays, of which **37.1%** meet the Sepsis-3 definition.

The clinical question is whether **routinely measured physiology that is *not* part of the sepsis definition** (lactate, white-cell count, heart rate, respiratory rate, temperature, and age) can separate septic from non-septic stays, and what different statistical methods reveal about the same data. Choosing predictors outside the SOFA score avoids circularity: the models recover the label from independent signals, not from the quantities used to construct it.

All inferential and modelling sections use the **36,529** stays with a complete set of these six measurements (50.0% Sepsis-3 positive). Because lactate is ordered preferentially in sicker patients, this complete-case subset is enriched for severity relative to the full cohort; this is revisited as a limitation in the conclusion.

The report follows a structured progression:

1. **Exploratory data analysis** (Section 2) describes the cohort and feature distributions by sepsis status.
2. **Hypothesis testing** (Section 3) quantifies which features differ between groups, focusing on effect size.
3. **Maximum likelihood estimation** (Section 4) models the typical lactate of a septic patient.
4. **Generalised linear models** (Section 5) determine which features carry independent predictive power.
5. **Kernel density estimation and mixture modelling** (Section 6) attempt to recover the two groups without labels.
6. **Metropolis-Hastings sampling** (Section 7.1) re-expresses the likelihood result in a Bayesian framework.
7. **A Hidden Markov Model** (Section 7.2) tests whether the ordering of observations carries extra structure.
8. **Predictive validation** (Section 8) subjects the model to out-of-sample cross-validation, calibration, precision-recall and decision-curve analysis, a non-linearity check, and a formal treatment of missing data.

2 Exploratory Data Analysis

This section examines the data before modelling, focusing on lactate, white-cell count, and the vital signs central to bedside recognition of sepsis.

In the modelling subset there are **18,255** non-septic and **18,274** septic stays (50.0% septic). The classes are imbalanced at roughly **1.0:1**, milder than many clinical-outcome datasets and not severe enough to require resampling; the inferential focus on effect size further reduces its impact.

Table 1: Summary statistics for the six features, by sepsis status.

Feature	Mean (no)	Mean (sep)	Med. (no)	Med. (sep)	SD (no)	SD (sep)
Age (years)	64.17	65.38	66.00	67.00	16.67	16.40
Heart rate, max (bpm)	102.59	108.89	99.00	106.00	76.68	50.90
Lactate, max (mmol/L)	2.58	3.10	2.10	2.20	2.26	2.87
Resp. rate, max (/min)	28.30	541.79	27.00	28.00	31.66	54637.16
Temperature, max (C)	37.25	37.56	37.17	37.39	0.74	0.97
WBC, max (K/uL)	14.05	16.06	12.80	14.20	9.32	13.59

Table 1 shows that septic stays tend to have higher lactate, higher white-cell counts, and faster heart and respiratory rates. These shifts match the bedside picture of sepsis, although the spread within each group is wide.

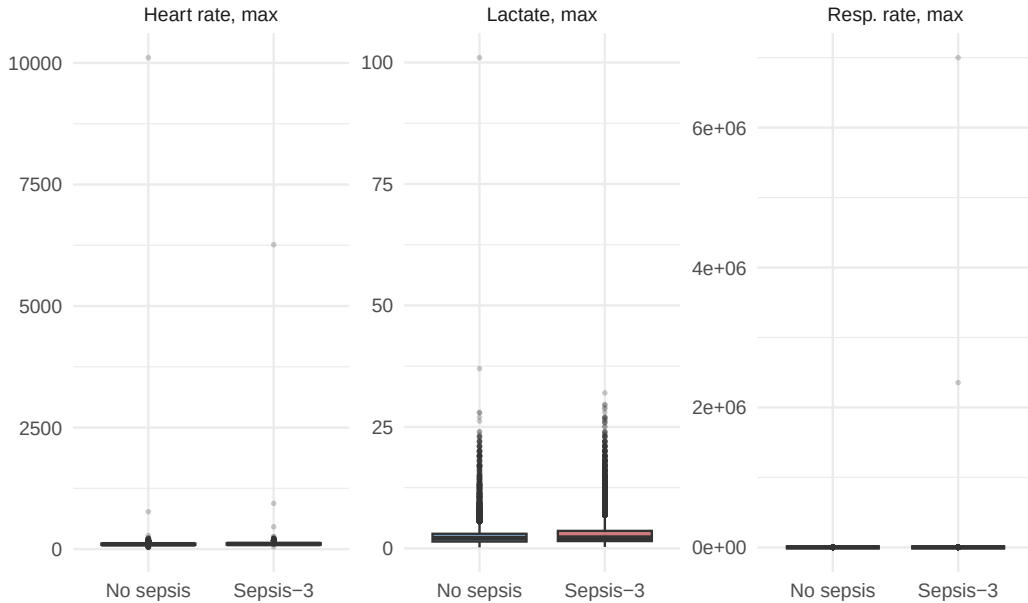


Figure 1: Three features by sepsis status. Septic stays show higher lactate, faster breathing, and faster heart rates, with substantial overlap.

Figure 1 confirms the direction of these differences but also the overlap: no single feature cleanly separates the groups, which motivates the multivariable model in Section 5.

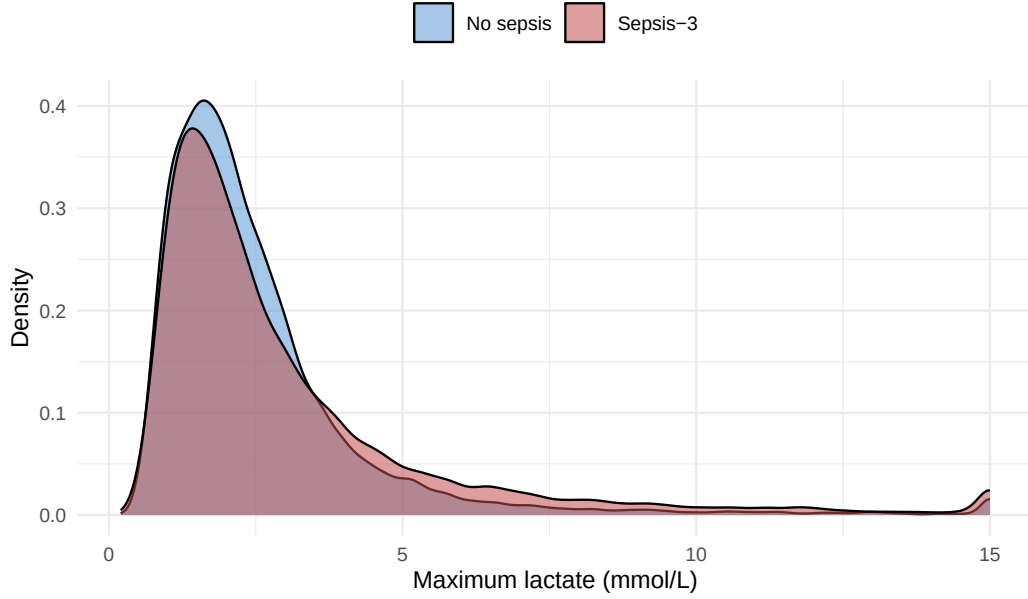


Figure 2: Distribution of maximum lactate by sepsis status (values capped at 15 mmol/L for display). Both groups are right-skewed; the septic distribution is shifted upward.

Figure 2 shows two overlapping, right-skewed distributions. The septic curve is shifted toward higher lactate, but the long right tail in both groups signals that a log transform will be useful for the distributional models in Sections 4 to 7.2.

Collectively, this section shows the data contain genuine, clinically coherent signal, but no single feature is sufficient for perfect separation.

3 Hypothesis Testing

Three features are compared between septic and non-septic stays, each with a different test. With **36,529** observations, even trivial differences yield very small p-values, so interpretation centres on **effect size** (the magnitude of the difference) rather than significance alone.

3.1 Test 1: Maximum lactate (Welch t-test)

The two groups have unequal variances, so a Welch t-test is appropriate.

H_0 : mean maximum lactate is equal in septic and non-septic stays

H_1 : mean maximum lactate differs between the groups

Table 2: Welch t-test: maximum lactate by sepsis status.

Statistic	Value
t	-19.18
df	34642
p-value	<1e-16
Mean (no sepsis)	2.58
Mean (Sepsis-3)	3.10
Mean difference	-0.52
Cohen's d	-0.20
95% CI for d	[-0.22, -0.18]

Table 2 shows mean maximum lactate of **2.58** mmol/L in non-septic stays versus **3.10** mmol/L in septic stays, a difference of **-0.52** mmol/L. The p-value is <1e-16, so the difference is not attributable to chance. Cohen's *d* of **-0.20** indicates a small-to-moderate effect: lactate separates the groups, but not perfectly.

3.2 Test 2: Maximum white-cell count (Wilcoxon rank-sum)

White-cell counts are heavily right-skewed, making a non-parametric Wilcoxon rank-sum test appropriate.

H_0 : the distribution of maximum WBC is the same in both groups

H_1 : the distributions differ

Table 3: Wilcoxon rank-sum: maximum WBC by sepsis status.

Statistic	Value
W	148,333,588
p-value	<1e-16
Rank-biserial r	-0.11
95% CI	[-0.12, -0.10]

Table 3 shows a significant difference in white-cell counts. The rank-biserial correlation of **-0.11** is small in magnitude, weaker than lactate and consistent with white-cell count being a noisier marker that rises in many non-septic stresses as well.

3.3 Test 3: Tachypnoea and sepsis (Chi-squared)

Respiratory rate is dichotomised at the **qSOFA** threshold of 22 breaths/min, a recognised marker of clinical deterioration. A chi-squared test of independence assesses the association with sepsis.

H_0 : tachypnoea ($\geq 22/\text{min}$) and sepsis are independent

H_1 : they are associated

Table 4: Chi-squared: tachypnoea (RR ≥ 22) by sepsis status.

Statistic	Value
Chi-squared	176.2
df	1
p-value	$<1e-16$
Cramer's V	0.07
95% CI	[0.06, 1.00]

Table 4 shows a significant association between tachypnoea and sepsis, with Cramer's V of **0.07** (a weak association). Dichotomising at a single threshold discards information; the GLM in Section 5 uses the continuous respiratory rate and recovers a stronger signal.

Collectively, all three features carry genuine signal. Lactate shows the largest effect; white-cell count and tachypnoea are weaker but real.

4 Maximum Likelihood Estimation

Lactate is strictly positive and right-skewed, fitting a **log-normal** model: the log of maximum lactate is assumed Normal. Maximum likelihood estimates the mean μ and standard deviation σ on the log scale, and the typical (median) lactate is recovered as e^μ .

Table 5: MLE for the log-normal model of septic maximum lactate.

Parameter	Estimate
mu (mean log-lactate sepsis)	0.865
sigma (SD log-lactate sepsis)	0.684
n (septic)	18,274
SE of mu	0.0051
Median lactate exp(mu) (mmol/L)	2.37
95% CI for median	[2.35, 2.40]

The maximum-likelihood estimate from Table 5 shows:

1. The typical (median) maximum lactate for a septic stay is about **2.37** mmol/L, above the 2 mmol/L threshold used in the septic-shock definition.
2. The log-scale standard deviation of **0.684** reflects the wide spread of lactate values.
3. The very small standard error shows the central value is estimated precisely, a direct consequence of the large septic sample.

5 Generalised Linear Models

The binary outcome calls for logistic regression. Two models are compared:

- **Model 1** uses maximum lactate only.
- **Model 2** adds white-cell count, heart rate, respiratory rate, temperature, and age.

A correlation check among the continuous predictors shows the strongest pairwise correlation is **0.06**, well below the 0.8 threshold that would signal redundancy, so all predictors are retained.

5.1 Logistic Regression Coefficients

Table 6: Logistic regression coefficients (log-odds scale).

Model	Term	Estimate	SE	p-value
M1: lactate only	(Intercept)	-0.238	0.016	<1e-16
M1: lactate only	lactate_max	0.085	0.005	<1e-16
M2: 6 predictors	(Intercept)	-21.578	0.604	<1e-16
M2: 6 predictors	lactate_max	0.078	0.005	<1e-16
M2: 6 predictors	wbc_max	0.018	0.001	<1e-16
M2: 6 predictors	hr_max	0.008	0.001	<1e-16
M2: 6 predictors	resp_rate_max	0.003	0.001	0.005
M2: 6 predictors	temp_max_c	0.522	0.016	<1e-16
M2: 6 predictors	age	0.011	0.001	<1e-16

Coefficients in Table 6 are on the log-odds scale. Converting to odds ratios aids interpretation.

Table 7: Model 2 odds ratios (per one-unit increase).

Term	Odds ratio	95% CI
lactate_max	1.081	[1.071, 1.091]
wbc_max	1.018	[1.015, 1.021]
hr_max	1.008	[1.007, 1.009]
resp_rate_max	1.003	[1.001, 1.006]
temp_max_c	1.685	[1.633, 1.739]
age	1.011	[1.009, 1.012]

The odds ratio for maximum lactate is **1.08**, meaning each additional 1 mmol/L multiplies the odds of sepsis by about 1.08, holding the other variables constant. The vital signs and age each contribute smaller but independent effects, with odds ratios above 1 indicating that faster heart and respiratory rates, higher temperature, and older age are each associated with a higher probability of sepsis once the others are accounted for.

5.2 Model Comparison

Table 8: Model comparison (lower AIC is better).

Model	AIC	Residual deviance	df
M1: lactate only	50257.1	50253.1	36527
M2: six predictors	48112.3	48098.3	36522

Table 8 shows the AIC falling from **50257** (lactate only) to **48112** (six predictors), a clear improvement: the additional features carry information about sepsis beyond lactate alone. The six-predictor model achieves an approximate AUC of **0.66**, indicating modest discrimination from physiology that is external to the sepsis definition.

Sepsis is therefore best explained by a combination of features rather than any single measurement.

6 Density Estimation and Mixture Models

Setting aside the sepsis labels, the question is whether maximum lactate alone splits the data into two natural groups.

6.1 Kernel Density Estimate

Silverman’s rule-of-thumb bandwidth is

$$h = 0.9 \times \min\left(s, \frac{IQR}{1.34}\right) \times n^{-1/5},$$

where s is the standard deviation, IQR the interquartile range, and n the sample size. Because lactate is right-skewed, the KDE is computed on the **log** scale, where the distribution is closer to symmetric.

The computed bandwidth is **0.068** on the log scale.

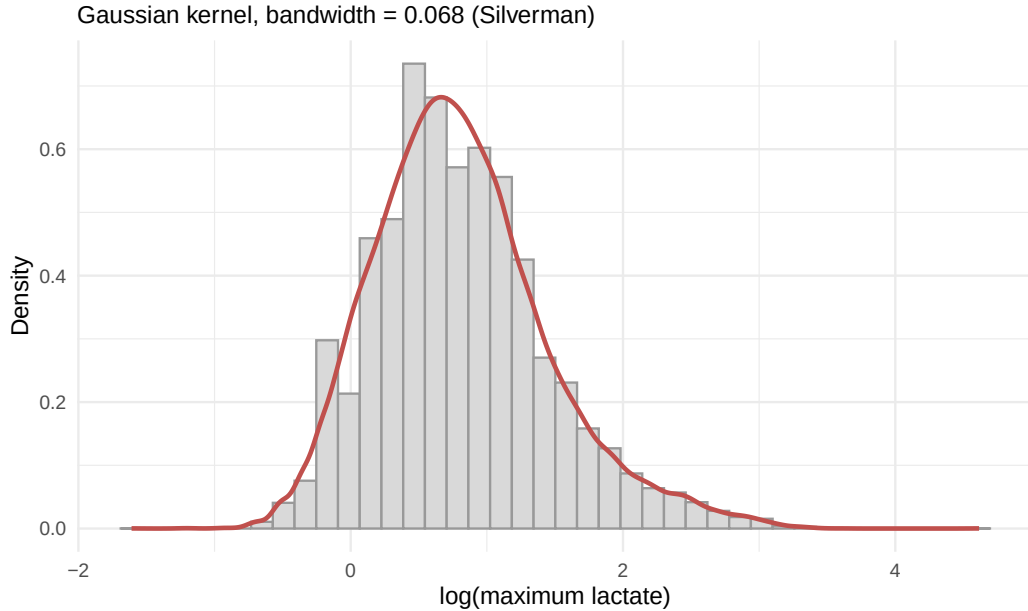


Figure 3: Kernel density estimate of log maximum lactate (Gaussian kernel, Silverman bandwidth). A shoulder above the main peak hints at a second, higher-lactate population.

Figure 3 shows a dominant peak with a right shoulder, suggesting the data may be a blend of a larger normal-lactate population and a smaller high-lactate one rather than a single homogeneous group.

6.2 Two-component Gaussian mixture

A two-component Gaussian mixture is fitted to log-lactate by Expectation-Maximisation, using no labels:

$$f(x) = \pi_1 \mathcal{N}(\mu_1, \sigma_1^2) + (1 - \pi_1) \mathcal{N}(\mu_2, \sigma_2^2).$$

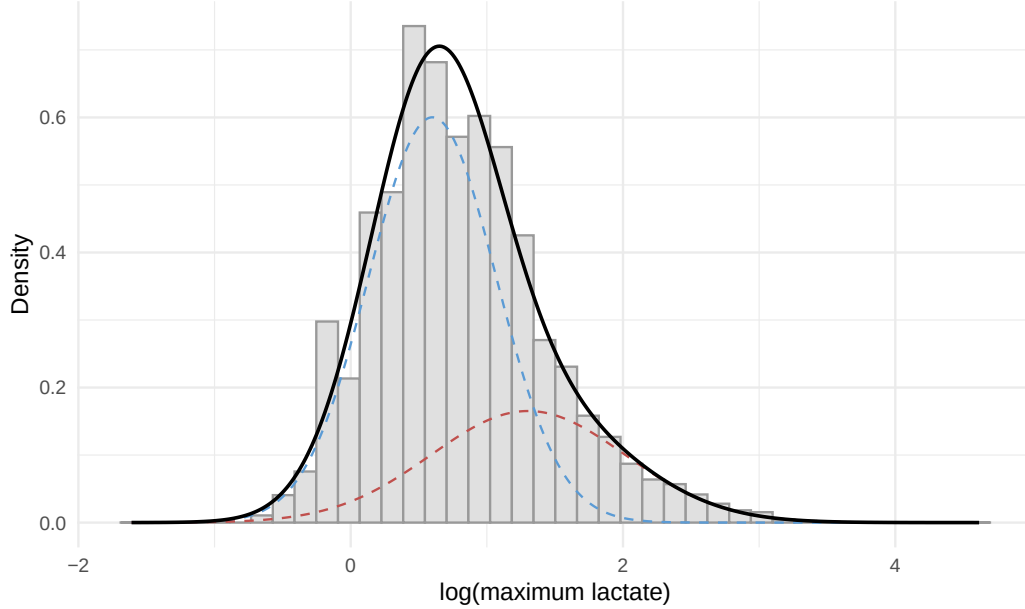


Figure 4: Two-component Gaussian mixture (dashed components, solid sum) fitted to log maximum lactate.

The mixture places one component near **1.82** mmol/L (back-transformed) and the other near **3.68** mmol/L, with mixing weights 0.70 and 0.30.

Matching the unlabelled components to the true labels gives an unsupervised agreement of **54.3%**. Unlike a marker such as tumour radius, which nearly separates its two classes, lactate alone only **partially** recovers sepsis status: the two populations overlap heavily, making a single biomarker a weak unsupervised classifier. This informative negative result illustrates why the Sepsis-3 definition combines infection with a multi-organ score rather than relying on any one number.

7 Bayesian Inference and MCMC

Maximum likelihood (Section 4) gave a single best estimate of the mean log-lactate. Bayesian inference returns a full posterior distribution, expressing uncertainty as a range of plausible values.

7.1 Part A: Bayesian inference via Metropolis-Hastings

7.1.1 Model setup

Log maximum lactate for septic stays is modelled as Normal with unknown mean μ and a known standard deviation fixed at the MLE $\hat{\sigma}$ from Section 4. A weak prior $\mu \sim \mathcal{N}(\log 2, 1^2)$ is centred at 2 mmol/L (the clinical threshold) but is wide enough to let the data dominate.

7.1.2 Likelihood and posterior

The posterior combines the Normal likelihood with the Normal prior,

$$p(\mu | x) \propto p(x | \mu) p(\mu),$$

which here is conjugate (and therefore Normal). The Metropolis-Hastings algorithm is nonetheless used to demonstrate a general sampling approach that does not rely on conjugacy.

Two chains are run from deliberately poor, opposite starting points and the first 2,000 iterations discarded as burn-in.

7.1.3 Convergence diagnostics

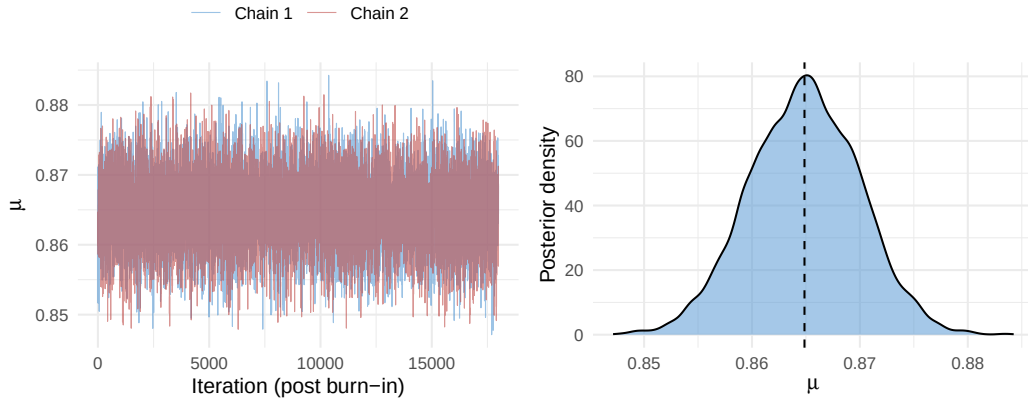


Figure 5: Trace plots (left) of two chains overlap after burn-in; posterior density (right) with the MLE marked (dashed).

The chains overlap after burn-in (Figure 5). The Gelman-Rubin statistic is $\hat{R} = 1.000$ (values below 1.05 indicate convergence) and the effective sample size is about **4,131**, so the posterior samples are reliable.

7.1.4 Posterior summary and comparison with MLE

Table 9: MLE versus Bayesian posterior for mean log-lactate.

Quantity	MLE	Bayes (M-H)
Mean log-lactate sepsis	0.865	0.865
SE / posterior SD	0.0051	0.0051
95% interval	[0.855, 0.875]	[0.855, 0.875]
Implied median lactate (mmol/L)	2.37	2.37

Table 9 shows the Bayesian and maximum-likelihood estimates are almost identical. With a large septic sample and a weak prior, the data dominate; a much stronger prior centred at log 2 would be needed to pull the estimate appreciably toward 2 mmol/L.

The two frameworks agree, but the Bayesian result additionally provides a full posterior, which becomes more valuable when samples are small or strong prior knowledge exists.

7.2 Part B: Hidden Markov Model

The mixture model treated stays as independent. A hidden Markov model allows consecutive stays to share a latent state via a transition matrix.

Note on ordering. The stays are not a true time series; their order is the processing order. The question is therefore deliberately narrow: *after splitting the data into two lactate states, does the ordering add any structure?* If not, the HMM should reproduce the mixture model, and this negative result is itself informative. The natural application of an HMM is to *within-patient* hourly vital-sign sequences, where a latent state genuinely evolves toward organ dysfunction; that temporal use is the subject of the follow-on longitudinal analysis (see the Conclusion) and is outside the scope of this cross-sectional report.

7.2.1 Choosing the number of states

Table 10: HMM model selection by BIC (lower is better).

States	Log-likelihood	BIC
1	-7835.3	15688.5
2	-7554.1	15171.1
3	-7544.5	15214.9

Table 10 selects the **2-state** model by BIC, consistent with the two-population structure seen in Section 6.

7.2.2 Emissions and transition matrix

Table 11: HMM emissions versus the Section 6 mixture components.

State	Emission mean (log)	Median lactate (mmol/L)	Mixture mean (Sec. 6)
State 1 (lower)	0.649	1.914	0.600
State 2 (higher)	1.629	5.096	1.303

Table 12: Estimated transition matrix.

From	To State 1	To State 2
State 1	0.822	0.178
State 2	0.818	0.182

Table 11 shows the HMM emission means almost coincide with the mixture components from Section 6.2: both methods place the two lactate populations in the same locations. The transition matrix (Table 12) has a stationary distribution of (0.82, 0.18). Any “stickiness” on the diagonal is the pattern EM tends to produce when fitting independent, overlapping data, not evidence of real sequential dependence.

7.2.3 Unsupervised classification accuracy

Decoding the most likely state for each stay and matching states to labels gives an unsupervised accuracy of **53.8%**, essentially the same as the mixture model (54.3%). The HMM learns nothing beyond the mixture: the two-lactate-population structure is a property of the **patients**, not of the order in which they were recorded.

8 Predictive Validation, Calibration and Missingness

The GLM in Section 5 reported an AUC computed **on the same data it was fitted to**, which is optimistically biased. A model intended for *real-time* prediction must be judged on data it has not seen, must be **calibrated** (predicted risks must match observed frequencies), and must be honest about the **missing-data** mechanism that shaped the analysis sample. This section addresses all three.

8.1 Out-of-sample discrimination

Because the cohort keeps only each patient’s first ICU stay, every patient contributes exactly one row, so an ordinary random split is already patient-disjoint (no leakage between train and test). Ten-fold cross-validation is used: each stay is scored by a model trained on the other nine folds.

Table 13: In-sample versus cross-validated discrimination.

Model	In-sample AUC	10-fold CV AUC	95% CI (CV, M2)
M1: lactate only	0.540	0.539	
M2: six predictors	0.661	0.661	[0.655, 0.666]

Table 13 shows the cross-validated AUC of the six-predictor model is **0.661** (95% CI 0.655 to 0.666), essentially unchanged from the in-sample **0.661**; with 36,529 observations and only six predictors the model does not overfit. A DeLong test confirms that the six-predictor model discriminates better than lactate alone (AUC 0.661 vs 0.539, $p < 1e-16$).

8.2 Calibration and clinical usefulness

Discrimination (ranking) is not the same as calibration (the numerical risk being correct). The cross-validated risks are grouped into deciles and the mean predicted risk compared with the observed sepsis rate.

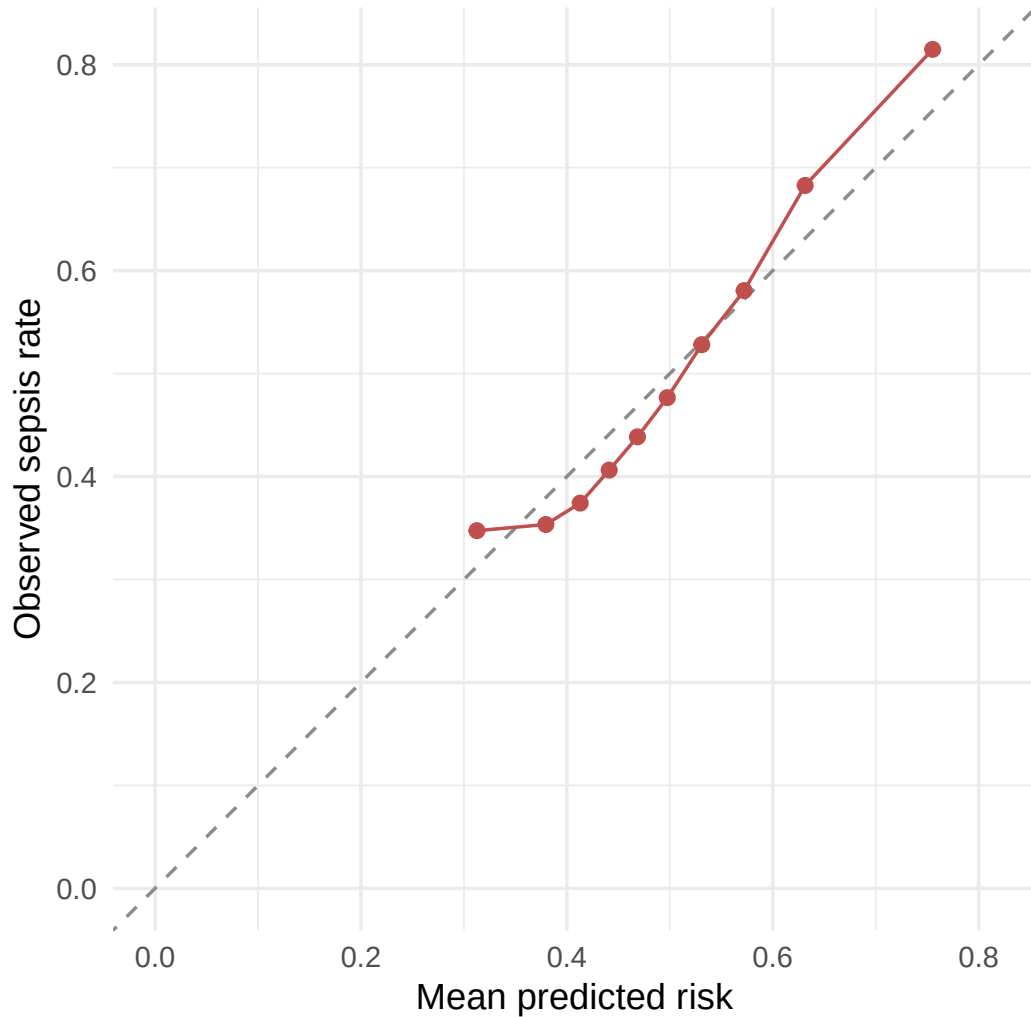


Figure 6: Calibration of the cross-validated six-predictor model (risk deciles). Points on the diagonal indicate predicted risks that match observed frequencies.

The **Brier score** is **0.231** (lower is better), against **0.250** for a null model that always predicts the prevalence, a 8% reduction. The **calibration slope** is **1.02** (1.0 is perfect); Figure 6 shows the decile points lie close to the diagonal, so the predicted probabilities are usable as risks, not merely as a ranking.

Because sepsis is imbalanced and the costs of missing it are asymmetric, ROC-AUC is supplemented with the **precision-recall AUC** and a **decision-curve (net-benefit)** analysis.

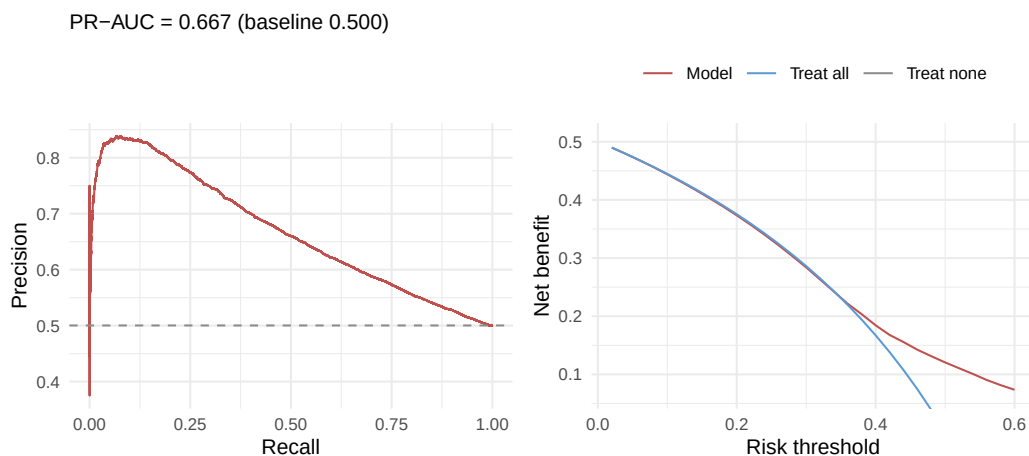


Figure 7: Left: precision-recall curve (dashed line = prevalence, the no-skill baseline). Right: decision curve showing net benefit of the model against treat-all and treat-none across risk thresholds.

The PR-AUC is **0.667** against a no-skill baseline of **0.500** (the prevalence), and Figure 7 shows the model yields positive net benefit above the treat-all and treat-none strategies across the clinically relevant range of thresholds, confirming that the discrimination translates into decision value.

8.3 Non-linearity of the lactate effect

The GLM assumed lactate acts **linearly on the log-odds** scale. A penalised thin-plate spline `s(lactate_max)` tests this assumption against a flexible alternative.

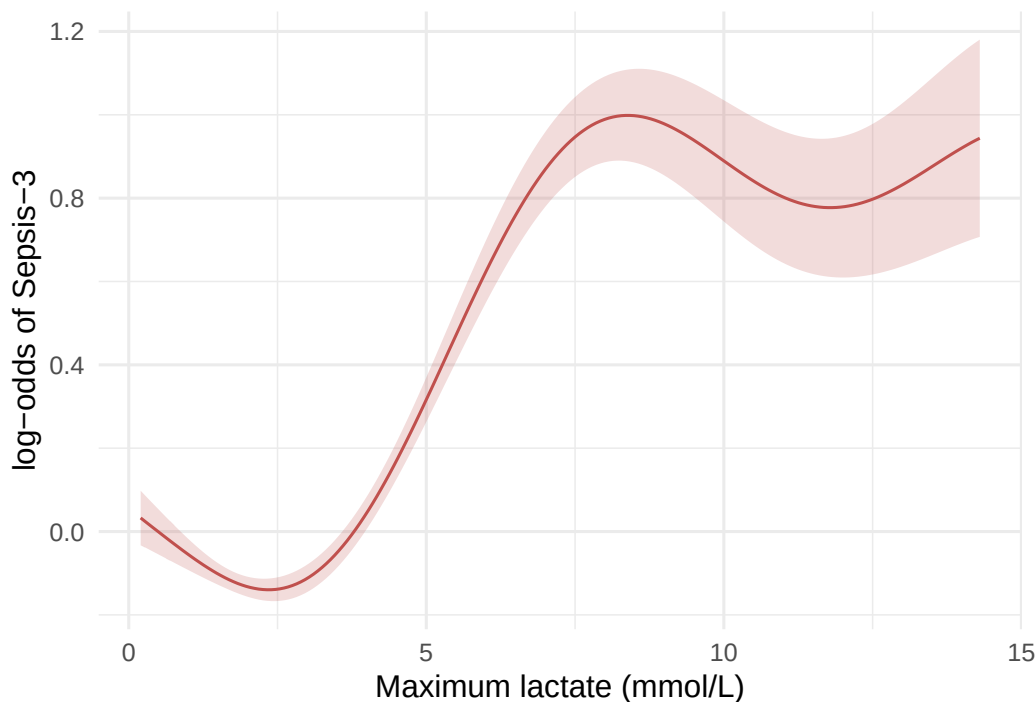


Figure 8: Penalised-spline estimate of the log-odds of sepsis as a function of maximum lactate (shaded 95% band). Curvature indicates a non-linear effect the linear GLM cannot capture.

The spline uses **7.2** effective degrees of freedom (1.0 would mean linear), and it lowers AIC by **223**

relative to the linear term, so the lactate effect is materially non-linear (Figure 8): risk rises steeply at low lactate and then flattens. A production model should therefore spline-transform lactate rather than enter it linearly.

8.4 Multicollinearity

Section 5 checked only three pairwise correlations. Variance-inflation factors provide a more complete assessment across all six predictors of Model 2.

Table 14: Variance-inflation factors (VIF > 5 signals collinearity).

Predictor	VIF
hr_max	1.14
temp_max_c	1.10
lactate_max	1.08
age	1.07
resp_rate_max	1.05
wbc_max	1.04

All VIFs in Table 14 are near 1 (maximum **1.14**), well below the usual cut-off of 5, so the coefficients are not destabilised by collinearity.

8.5 The missing-data mechanism

Every inferential section so far used only the 36,529 stays with a **complete** set of the six features, chiefly limited by lactate, which is *ordered at the clinician’s discretion*. Whether a lactate exists is therefore not random: it is itself a marker of concern.

Lactate is missing in **40.9%** of the full cohort. Sepsis-3 prevalence is **49.8%** among stays with a measured lactate versus **18.7%** among those without (chi-squared $p < 1e-16$), so *the act of measuring lactate is itself strongly informative*, and complete-case analysis enriches the sample for severity. Missingness is plausibly **MAR given the recorded vitals** (sicker-looking patients get labs), but a **MNAR** component cannot be excluded.

As a sensitivity analysis the six predictors are **multiply imputed** (MICE, predictive mean matching) on the full cohort and the logistic model re-fitted, pooling coefficients by Rubin’s rules.

Table 15: Odds ratios: complete-case versus multiply-imputed full cohort.

	Predictor	OR (complete-case)	OR (imputed cohort)	95% CI (imputed)
2	lactate_max	1.081	1.089	[1.079, 1.100]
3	wbc_max	1.018	1.034	[1.031, 1.038]
4	hr_max	1.008	1.011	[1.010, 1.012]
5	resp_rate_max	1.003	1.000	[1.000, 1.000]
6	temp_max_c	1.685	1.954	[1.902, 2.007]
7	age	1.011	1.013	[1.012, 1.014]

Table 15 shows the odds ratios keep the same direction and comparable magnitude after imputing the

full cohort: the associations are not an artefact of the complete-case restriction, though the absolute sepsis prevalence differs between the samples. This missingness signal is itself a lesson for the real-time model: *whether and when a test is ordered* is a usable predictor, not just a nuisance.

Collectively, this section shows the six-predictor model generalises out of sample, is well calibrated and clinically useful, benefits from a non-linear lactate term, is free of collinearity, and is robust to the missing-data mechanism, while also revealing that measurement itself carries information.

9 Conclusion

This report asked whether routine physiology that is *not* part of the sepsis definition can identify Sepsis-3, and what different statistical methods reveal about the same data.

- **Hypothesis tests** (Section 3) found genuine differences between septic and non-septic stays, largest for lactate (Cohen’s $d = -0.20$), and weaker but real for white-cell count and tachypnoea.
- The **GLM** (Section 5) showed lactate is the single strongest predictor, while vital signs and age add independent information: AIC fell from 50257 to 48112, with an approximate AUC of 0.66.
- The **mixture model** (Section 6.2) and **hidden Markov model** (Section 7.2) both recovered two lactate populations without labels, agreeing almost exactly with each other (54.3% vs 53.8%). Crucially, lactate alone only partially recovers sepsis, a reminder that the diagnosis is multi-dimensional.
- The **MLE** (Section 4) and **MCMC** (Section 7.1) agreed closely on the typical septic lactate (≈ 2.37 mmol/L), with nearly identical interval estimates.
- **Validation** (Section 8) confirmed the six-predictor model holds up out of sample (10-fold CV AUC 0.661, DeLong $p < 1e-16$ versus lactate alone), is well calibrated (Brier 0.231, slope 1.02), gains from a non-linear lactate term, and keeps its associations under multiple imputation, while also showing that *whether* a lactate was ordered is itself predictive.

The methodological message echoes across tools: several independent methods converge on the same, clinically coherent picture. No single physiological variable identifies sepsis on its own, which is precisely why Sepsis-3 couples suspected infection with a multi-organ SOFA score. The geometry of one biomarker is not enough; it takes a combination of signals to recognise a dysregulated response to infection.

Limitations and the path to real-time prediction. The analysis subset is restricted to stays with a measured lactate, which enriches for severity (quantified and partly addressed by the multiple-imputation sensitivity analysis in Section 8.5); sepsis was operationalised with a first-24h (“day-1”) SOFA and a small set of documented simplifications (Section 05 header); and the ordering-based HMM is illustrative, since stays are not a true time series. Most importantly, this report is **cross-sectional**: it collapses each stay into one row of first-24h summaries and answers “*was this stay septic?*” rather than the project’s ultimate question, “*will this patient become septic in the next few hours, given the data so far?*” That real-time question requires an **hourly patient panel**, a defined **sepsis-onset time** and prediction horizon, **survival / time-varying-hazard** modelling of time to onset, a **within-patient** (not ordering-based) HMM, and **time-dependent, leakage-safe** evaluation benchmarked against streaming clinical scores (qSOFA, SIRS, hourly SOFA, MEWS). Building that longitudinal foundation is the subject of the follow-on work, for which the associations, effect sizes, non-linearities, and missingness structure established here provide the modelling priors.

Acknowledgment

During this project, I applied the **PAIR framework** to responsibly use generative AI alongside primary sources.

- **Problem:** Build a reproducible MIMIC-IV Sepsis-3 pipeline and a layered statistical analysis (EDA, hypothesis testing, MLE, GLMs, density and mixture modelling, Metropolis-Hastings sampling, and a hidden Markov model) in R, supported by a coherent written report.
- **AI:** I used AI assistants to discuss implementation choices, clarify statistical concepts, and refine the writing.
- **Interaction:** All AI-generated suggestions were critically reviewed, checked against primary sources (Singer et al., 2016; Seymour et al., 2016; Vincent et al., 1996; Silverman, 1986; Robert & Casella, 2004; Gelman et al., 2013; Rabiner, 1989; Visser & Speekenbrink, 2010), and adapted to the structure and seed-based reproducibility of this work.
- **Reflection:** AI assistance accelerated drafting and improved clarity, while the interpretation, code, model choices, and conclusions remain my own.

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